First Duplicate Value

AlgoExpert · Medium · Arrays

The problem

- Values are in [1, n]; return the value whose second occurrence comes first, else -1.
- Mark visited values by negating the slot at index |v|-1 (mutates input).

Intuition

- Values are between one and n, so each value can mark its own index by flipping that slot negative.

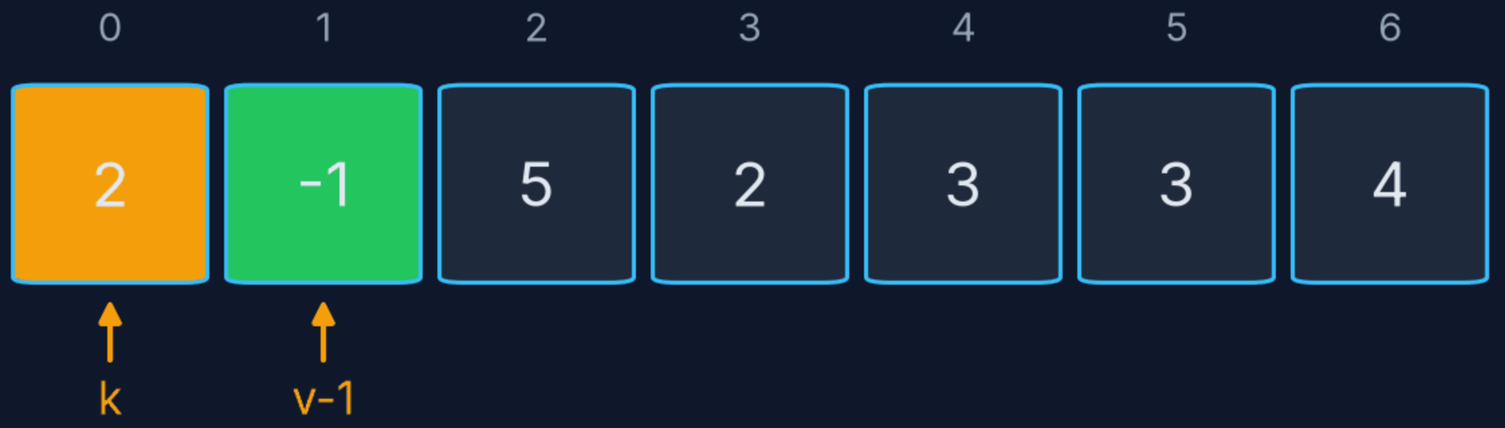
Brute force vs optimal

- Brute force: for each element, scan ahead → $O(n^2)$
- Hash set of seen values → O(n) time, O(n) space
- Values are 1…n → mark by negating a[v−1] → O(1) space

Approach

- 1. For each value, look at index abs value minus one
- 2. If that slot is already negative, we've seen it: return it
- 3. Otherwise flip it negative

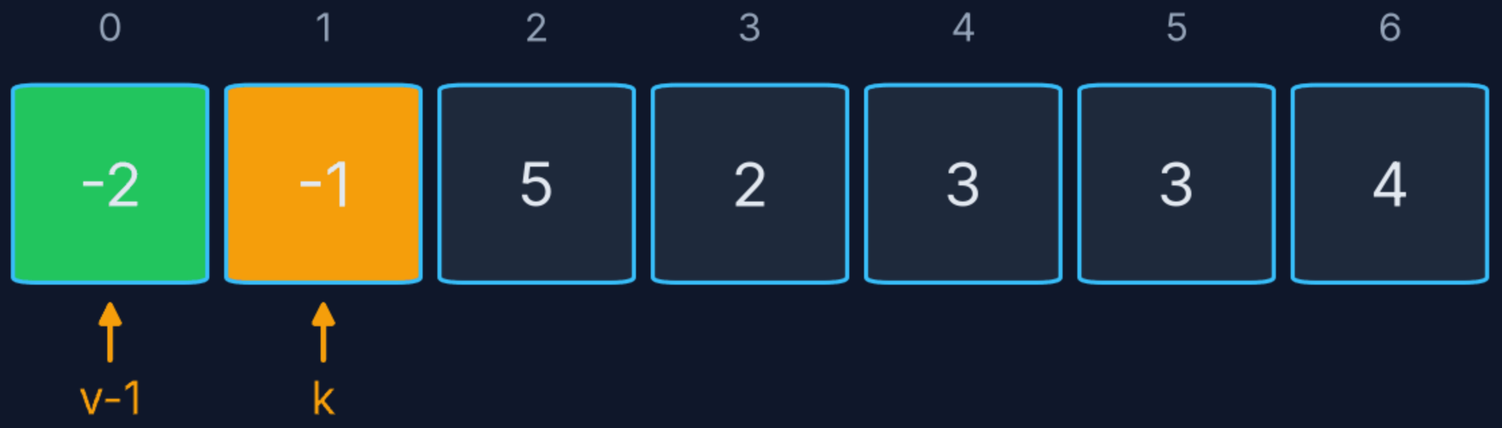
Worked example




```
value 2: flip slot 1 negative
```

Mark 2 as seen by negating index 1

Worked example




```
value 1: flip slot 0 negative
```

Mark 1 as seen by negating index 0

Worked example




```
value 5: flip slot 4 negative
```

Mark 5 as seen by negating index 4

Worked example




```
value 2: slot 1 already negative → duplicate!
```

Answer: 2

Java solution

```java
public class FirstDuplicateValue {
    public static int firstDuplicateValue(int[] a) {
        for (int x : a) {
            int v = Math.abs(x);
            if (a[v - 1] < 0) return v;
            a[v - 1] = -a[v - 1];
        }
        return -1;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Use abs() — the slot may already be negated
- No duplicate → −1
- Mutates the input; mention or restore it

Interview tips & follow-ups

- Ask whether modifying the input is allowed
- Follow-up: read-only array → Floyd's cycle detection

Recap

- Values are between one and n, so each value can mark its own index by flipping that slot negative.
- Complexity: O(n)